

```

// JUNAYED HASAN
#include <bits/stdc++.h>
using namespace std;
int main(){
    int n;
    cout << "----1st LAB: Right Shift The Array Value And
    Sorting----\n";
    cout << "Enter array size: "; cin >> n; int arr[n];
    // Array Input
    cout << "Enter array elements: ";
    for (int i = 0; i < n; i++){cin >> arr[i];}
    int position, value;
    cout << "Enter Index: "; cin >> position;

    // Error Handling
    if (position < 0 || position > n){
        cout << "Invalid Index! Try Again\n\n"; return main();
    }
    cout << "Enter value: "; cin >> value;

    // Right shift
    for (int i = n; i > position; i--){arr[i] = arr[i - 1];}
    arr[position] = value; n++;
    // Output right shifted array

    cout << "Right shifted values: ";
    for (int i = 0; i < n; i++){cout << arr[i] << " ";}
    cout << endl;

    // Bubble Sort
    for (int i = 0; i < n - 1; i++){
        for (int j = 0; j < n - i - 1; j++){
            if (arr[j] > arr[j +
1]){
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] =
temp;}}}

    // Output sorted array
    cout << "Sorted values: ";
    for (int i = 0; i < n; i++){
        cout << arr[i] << " ";
    }
    cout << endl;

```

```

"C:\projects\programming He X
----1st LAB: Right Shift The Array Value And Sorting----
Enter array size: 5
Enter array elements: 5 4 1 2 3
Enter Index: 10
Invalid Index! Try Again

----1st LAB: Right Shift The Array Value And Sorting----
Enter array size: 5
Enter array elements: 5 4 1 2 3
Enter Index: 2
Enter value: 33
Right shifted values: 5 4 33 1 2 3
Sorted values: 1 2 3 4 5 33

Process returned 0 (0x0)   execution time : 36.065 s
Press any key to continue.

```

```

        return 0;
    }
    // JUNAYED HASAN
    #include <bits/stdc++.h>
    using namespace std;
    int main(){
        string s;
        cout << "----2nd LAB: Tokenization---\n\n";
        cout << "Enter a string: ";
        getline(cin, s);
        int i = 0;
        while (i < s.length()){
            while (i < s.length() &&
                (s[i] == ' ' ||
                 s[i] == '@' ||
                 s[i] == ',' ||
                 s[i] == ';' ||
                 s[i] == '$' ||
                 s[i] == ':')){
                i++;
            }

            if (i >= s.length())
                break;
            cout << "<";
            while (i < s.length() &&
                s[i] != ' ' &&
                s[i] != '@' &&
                s[i] != ',' &&
                s[i] != ';' &&
                s[i] != '$' &&
                s[i] != ':'){
                cout << s[i];
                i++;
            }
            cout << ">" << endl;
        }
        return 0;
    }

```



The screenshot shows a Windows command prompt window titled "C:\projects\programming He X". The output of the program is as follows:

```

----2nd LAB: Tokenization---

Enter a string: @@@Junayed Hasan CSE,Compiler@Design;BUB
T$Dhaka:Bangladesh;;Saddam@Hossain,2026$$$ Test@@End:::
<Junayed>
<Hasan>
<CSE>
<Compiler>
<Design>
<BUBT>
<Dhaka>
<Bangladesh>
<Saddam>
<Hossain>
<2026>
<Test>
<End>

Process returned 0 (0x0)   execution time : 13.961 s
Press any key to continue.

```

```

// JUNAYED HASAN
#include <bits/stdc++.h>
using namespace std;
string isValidIdentifier(string s) {
    string keywords[] = {
        "auto", "break", "case", "char", "const", "continue",
        "default", "do", "double", "else", "enum", "extern",
        "float", "for", "goto", "if", "int", "long",
        "register", "return", "short", "signed", "sizeof",
        "static", "struct", "switch", "typedef", "union",
        "unsigned", "void", "volatile", "while", "class",
        "public", "private", "protected", "this", "new",
        "delete", "try", "catch", "throw", "bool", "true",
        "false", "using", "namespace", "virtual", "friend",
        "inline", "operator", "template", "typename"
    };
    if (s.empty()) {return "Invalid Identifier";}
    for (int i = 0; i < 56; i++) {
        if (s == keywords[i]) {
            return "Invalid Identifier";
        }
    }
    if (!(s[0] >= 'A' && s[0] <= 'Z') ||
        (s[0] >= 'a' && s[0] <= 'z') ||
        s[0] == '_')) {
        return "Invalid Identifier";
    }
    for (int i = 1; i < s.size(); i++) {
        if (!(s[i] >= 'A' && s[i] <=
'Z') ||
            (s[i] >= 'a' && s[i] <= 'z')
||
            (s[i] >= '0' && s[i] <= '9')
||
            s[i] == '_')) {
            return "Invalid Identifier";}}
    return "Valid Identifier";
}
int main() {
    string s;
    cout<<"---3rd LAB: Identifier_Checker---\n\n";
    cout<<"\nEnter a identifier name: ";
    getline(cin, s);

```

```

Valid Identifier
---3rd LAB: Identifier_Checker---
Enter a identifier name: abc_
Valid Identifier
---3rd LAB: Identifier_Checker---
Enter a identifier name: abc123
Valid Identifier
---3rd LAB: Identifier_Checker---
Enter a identifier name: _123
Valid Identifier
---3rd LAB: Identifier_Checker---
Enter a identifier name: 123abc
Invalid Identifier
---3rd LAB: Identifier_Checker---
Enter a identifier name: 123
Invalid Identifier
---3rd LAB: Identifier_Checker---
Enter a identifier name: abc@123
Invalid Identifier
---3rd LAB: Identifier_Checker---
Enter a identifier name: hello world'
Invalid Identifier
---3rd LAB: Identifier_Checker---
Enter a identifier name: int
Invalid Identifier
---3rd LAB: Identifier_Checker---
Enter a identifier name: _
Valid Identifier
---3rd LAB: Identifier_Checker---
Enter a identifier name:

```

```
cout<<endl;  
cout<<isValidIdentifier(s);  
main(); return 0;}
```